

Class 39: Proof Review

Review: What two things must be shown to prove T is a linear transformation from vector space V to vector space W ? What are the “Let” and the “Show” statements for each?

Review: What three things must be shown to prove W is a subspace of V ?

Connections

If A is any $m \times n$ matrix then $T(\mathbf{x}) = A\mathbf{x}$ is a linear transformation.

How is the range of T related to A ?

How is the kernel of T related to A ?

What is the domain space and the codomain space for this transformation?

If A has a pivot in every column, what can you say about T ?

If A has a pivot in every row, what can you say about T ?

How do you construct the standard matrix A for a given T ?

If $\mathbf{x}$ is an eigenvector for A with eigenvalue 3, what is $T(\mathbf{x})$?

Practice: If $T\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $T\left(\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$ what is $T\left(\begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}\right)$

Practice: Let V be the set of all vectors $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ in $\mathbf{R}^3$ such

that $a + b + c = 0$. Prove that V is a subspace of $\mathbf{R}^3$ and find a basis for V .